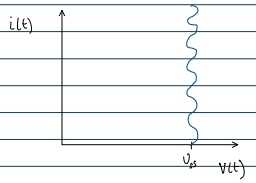
Electrochemical Impedance Spectroscopy (EIS)

- Apply a small amplitude sinusoidal potential @ various frequencies (+ analyze current response)
- EIS can be used to interpret everything that affects flow of current in echem system:
  - + Electrical resistance in current path
  - + Interfacial capacitance
  - + MT of reactants
  - + Kinetics of relevant hetero/homog. rxns.
- But system/measurement is complex + it can be easy to over-interpret results.
- EIS measurement:
  - + Apply small amplitude (~10mV) oscillating potential around some S.S. potential (V_{ss})



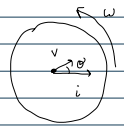
Potential represented w/ cos function

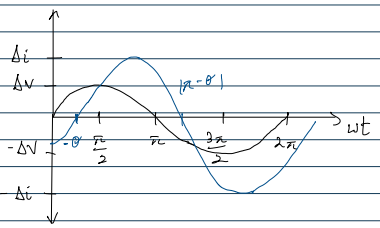
$V(t) = V_{ss} + \Delta V \cos(\omega t)$ (ω: angular frequency; ΔV: amplitude)

* How does i(t) vary w/ V(t)

- w/ small amplitude, we can expect linear response but might have some phase shift

$i(t) = i_{ss} + \Delta i \cos(\omega t + \theta)$ (θ: phase angle)





We can imagine current response to potential is affected by real electrochem processes that can be modelled by resistances (eg. C-t/kinetics) and capacitances (eg. double-layer charging)

+ Resistor: Ohm's law:

$$i(t) = \frac{V(t)}{R} = \frac{\Delta V}{R}\cos(\omega t) \rightarrow \text{phase angle } (\theta = 0)$$



+ Capacitor: $q = CV$

$$i(t) = C\frac{dV(t)}{dt} = -\omega C \Delta V \sin(\omega t)$$

$$= \omega C \Delta V \cos\left(\omega t - \frac{\pi}{2}\right)$$

↳ phase angle $= \frac{\pi}{2}$



Its convinient to express relationship b/w i-V as complex numbers

- Express y-component as imaginary by multiplying by $\sqrt{-1}$ (j) and x-component as real, then:

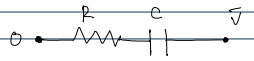
$\tilde{V} = \tilde{i}\tilde{Z}$ → must have units of resistance.

↳ express y-component as imaginary by multiplying by $\sqrt{-1}$ (j) and x-component as real, then:

+ capacitor : $\bar{i} \rightarrow$, $\bar{v}\downarrow$ $\qquad \bar{V} = -j\bar{i}\,\frac{1}{\omega C}$ (must have units of resistance; capacitive resistance; $\frac{1}{\omega C}$ = derivative)

+ Resistor : $\bar{v} \rightarrow \bar{i}$ $\qquad \bar{V} = R\bar{i}$

Ex) If we have a resistor + capacitor in series w/ voltage $\bar{V}$ applied across them



↳ Total voltage must equal sum of voltage across each component

$$\bar{V} = \bar{V}_R + \bar{V}_C$$

$$= R\bar{i} - j\bar{i}\,\frac{1}{\omega C} = \bar{i}\left(R - \frac{j}{\omega C}\right) = \bar{i}\,\bar{Z}$$

this quantity we can define as complex impedence → Resistive contributions from both capacitors + resistors

$$\boxed{\bar{Z} = Z_{Re} + j\,Z_{Im} = \frac{\bar{V}}{\bar{i}}}$$

How does this help us understand e-chem system?

ex) Consider irreversible rxn w/ Tafel kinetics

+ current from double layer

$$i(t) = i_o \exp\left[\frac{\alpha_a F}{RT}(V - \phi_o)\right] + C_{dl}\frac{\partial(V-\phi)}{\partial t}$$

Apply Euler's formula ($\exp(j\omega t) = \cos(\omega t) + j\sin(\omega t)$)

Plug

$$\begin{cases} V(t) = V_{ss} + \Delta V\cos(\omega t) \Rightarrow V_{ss} + Re\left(\bar{V}\exp(j\omega t)\right) \\ i(t) = i_{ss} + \Delta i\cos(\omega t) \Rightarrow i_{ss} + Re\left(\bar{i}\exp(j\omega t)\right) \end{cases}$$

+ Assume $\phi_o = i R_\Omega$ (ohmic drop)

+ Assume we can linearize exponentials

$$\bar{i} = \left(\frac{\alpha_a F i_{ss}}{RT} + j\omega C_{dl}\right)\left(\bar{V} - \bar{i}R\right)$$

Rearrange to get $\bar{Z} = \frac{\bar{V}}{\bar{i}}$; $\qquad \frac{\bar{i}}{\bar{V} - iR_\Omega} = \left(\frac{\alpha_a F i_{ss}}{RT} + j\omega C_{dl}\right)$

$$\frac{\bar{V}}{\bar{i}} = R_\Omega + \left(\frac{\alpha_a F i_{ss}}{RT} + j\omega C_{dl}\right)^{-1} = \bar{Z}$$

$$Z^t = \frac{R_\Omega + \frac{\alpha_a F i_{ss}}{RT}}{\left(\frac{\alpha_a F i_{ss}}{RT}\right)^2 + (\omega C_{dl})^2} + \frac{-j\omega C_{dl}}{\left(\frac{\alpha_a F i_{ss}}{RT}\right)^2 + (\omega C_{dl})^2}$$

$$Z^{\dagger} = \frac{R_{\Omega} + \frac{\alpha_a F i_{ss}}{RT}}{\left(\frac{\alpha_a F i_{ss}}{RT}\right)^2 + (\omega C_{dl})^2} + \frac{-j\omega C_{dl}}{\left(\frac{\alpha_a F i_{ss}}{RT}\right)^2 + (\omega C_{dl})^2}$$

Z_{Re} + $j \cdot Z_{im}$

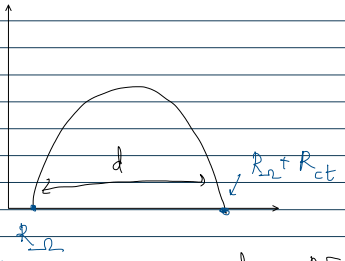
A common way to analyze EIS is to make a Nyquist Plot (Plot Z_{im} vs. Z_{Real})

Impedence: resistance to flow of current (units of ohms)

(ohmic drop)
Solution resistance

@ $\omega \to \infty$; $Z_{Re} = R_s$

Diameter $\propto \frac{RT}{\alpha_a F i_{ss}}$



(Solution resistance)

$d \propto \frac{RT}{\alpha_a F i_{ss}}$

As $d \uparrow$, $R_{ct} \uparrow$

$\omega \to \infty$ (R_Ω)
$\omega \to 0$ (

} Do analyses via the